

REFRACTION

Snell's law (*law of refraction*) describes the path of light rays passing the boundary between two different media, e.g. from air into glass. It is essential for the understanding of important applications in optics, e.g. lenses used in glasses or optical devices like cameras.

In this lab you are going to investigate the refraction of light rays when they strike the interface between two transparent media.

MEASUREMENTS

REFRACTION ON RECTANGULAR PRISM

SETUP:

Place the rectangular prism on a sheet of paper. Aim the light ray at one of the long sides of the rectangular prism such that it leaves the prism through its opposite side. Copy the incoming and outgoing light rays to the paper. Repeat this for a total of at least eight different angles. Measure the angles of the incoming and the refracted light rays, and the parallel translation between the incoming and the outgoing rays (with error estimates).

Dimensions of prism:

α											$\pm$
β											$\pm$
d (mm)											$\pm$

ANALYSIS

- Graph the sine of the incoming angle vs. the sine of the refractive angle and fit the expected function to the data. Determine the refractive index of the glass prism from the fit parameter (refractive index for air $n_A \approx 1$).
- Graph the refractive index vs. the incoming angle and fit the expected function to the data. Compare the fit parameter to the one in a).
- Add residual plots to both diagrams and discuss the quality of the fits.
- Derive a formal expression for the parallel displacement as a function of the incoming angle α , the refractive index n , and the width w of the glass prism. Calculate the predicted displacement (with its absolute uncertainty) for at least one data point and compare the result with the measured value.

ADDITIONAL TASKS

1. Graph the displacement vs. the incoming angle and fit the expected function to the data with the refractive index as fit parameter.
2. According to *Fermat's principle*, light always takes the path between two given points that takes the least time. Show that for a light ray going from a point in a first medium to a point in a second medium this is equivalent to Snell's law. The idea is to derive a function to calculate the time it takes light to travel from the first to the second point as a function of the point on the boundary between the two media, and to show that at the minimum of this function Snell's law is fulfilled.
3. Derive a formal expression for the deflection of a light ray that enters the prism through its short side and leaves through the long side. Visualise some typical examples.
4. Simulate the path of a light ray through a medium with a continuously changing refractive index. To do this, write a Python program that divides the medium into thin layers and calculates the successive refractions between neighbouring layers (assumed to have a constant refractive index). Visualise a few rays for different cases, i.e. a linear increase in the refractive index. Can you find a situation where light initially travelling downwards is reversed (this is the explanation for the phenomenon known as *mirage*).